Deploy a Python API

using FastAPI or Flask with
Gunicorn + NGINX

1. Install Python

```
sudo apt update && sudo apt upgrade -y

sudo apt install python3 python3-pip python3-venv –y

mkdir <project_name>

cd <project_name>


pip install fastapi "uvicorn[standard]”


nano main.py
```

2. Decide what you want to do

Main goal : Run some Fortran 72

Move your folder to the VM :

```
scp -r <src_path> <vm_user>@<vm_ip>:<destination_path>
cd <destination_path>
sudo chown -R $USER:$USER .
```

Compile and execute the project :

```
gfortran -o <executable_name> -ffixed-form -std=legacy <dependencies>
```

```
./<executable>
```

3. Open the communication port

DEMONSTRATION

4. Python API code

1. Copy the executable to the temporary folder

```
shutil.copy(EXECUTABLE_PATH, temp_dir)
```

2. Run the Fortran program

```
result = subprocess.run(["./ms"], cwd=temp_dir, capture_output=True,
text=True, check=True)
```

3. Create the ZIP archive containing the results (.lis and others)

```
for file in files:
  if file != "ms":
    file_path = os.path.join(root, file)
    zipf.write(file_path, arcname=file)
```

4. Send the file and schedule its deletion immediately after

```
background_tasks.add_task(cleanup_file, zip_path)
```

5. Return

```
return FileResponse( path=zip_path, filename=zip_filename,
media_type='application/zip')
```

5. Launch the API

VM :

cd exec-fortran

uvicorn main:app --reload --host 0.0.0.0

Browser :

http://<VM_Public_IP>:8000/run-simulation